



#KubeCon #CloudNativeCon
#OpenInfraSummit #PyTorchCon

China 2026

What We Learned Securing AI Agents at Scale on Multi-Tenant Kubernetes Clusters

Pengfei Ni · Microsoft Azure



01

Why an Agent Runtime



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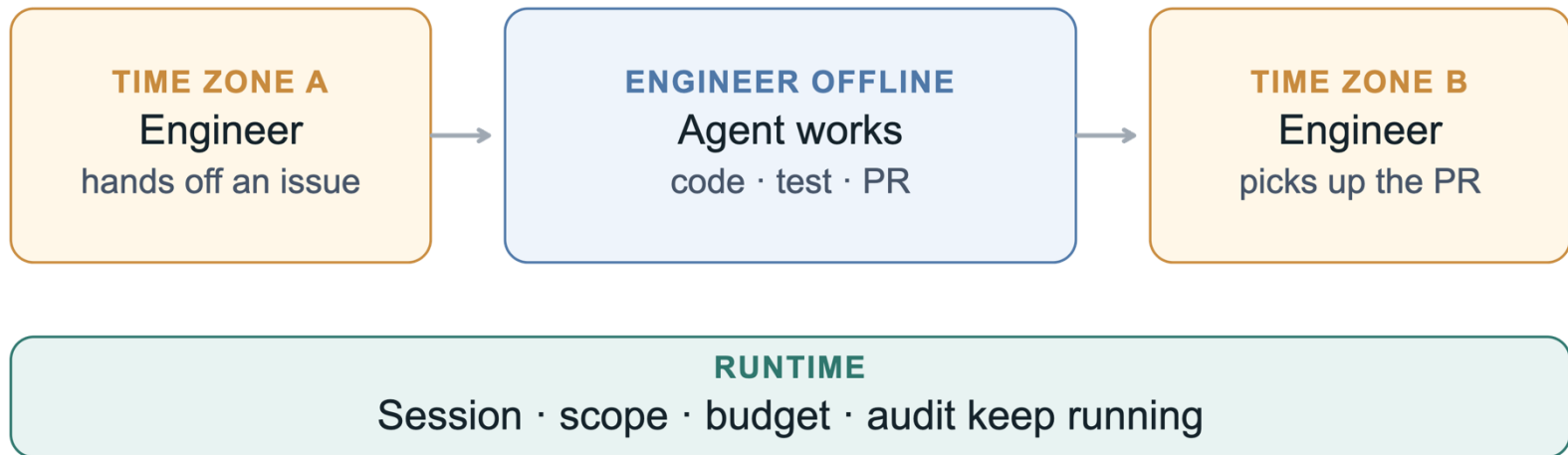


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Work outlives the engineer



Harness vs Runtime

Agent Harness

understands the task · picks tools · acts

GitHub Copilot

Hermes

OpenClaw

your own



Agent Runtime

runs untrusted agents isolated, safe and long-lived

Sandbox

isolated execution

Controlled access

gateway · broker

Skills & memory

what enters the context

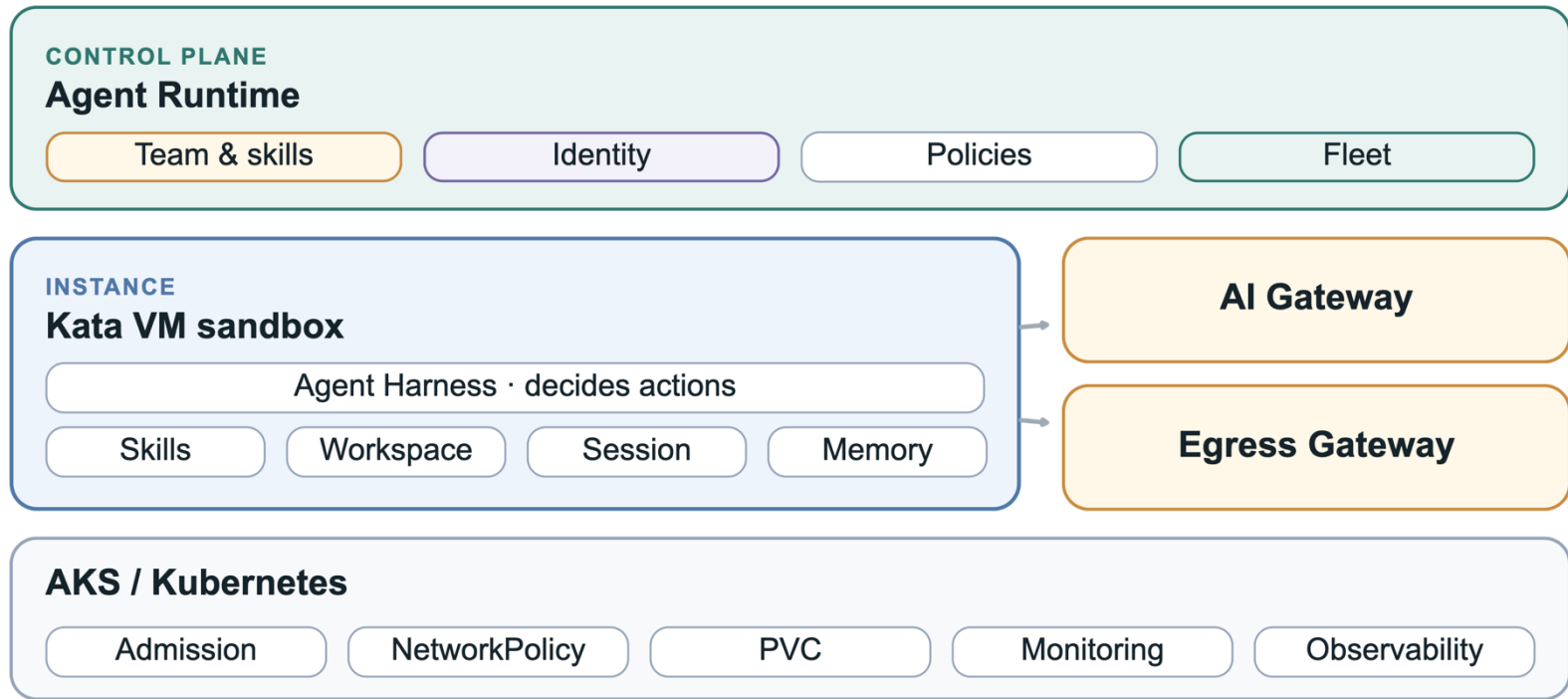
Control & govern

lifecycle · quota · cost · audit

Trust & security

Agent ID · short-lived credentials

Agent Runtime

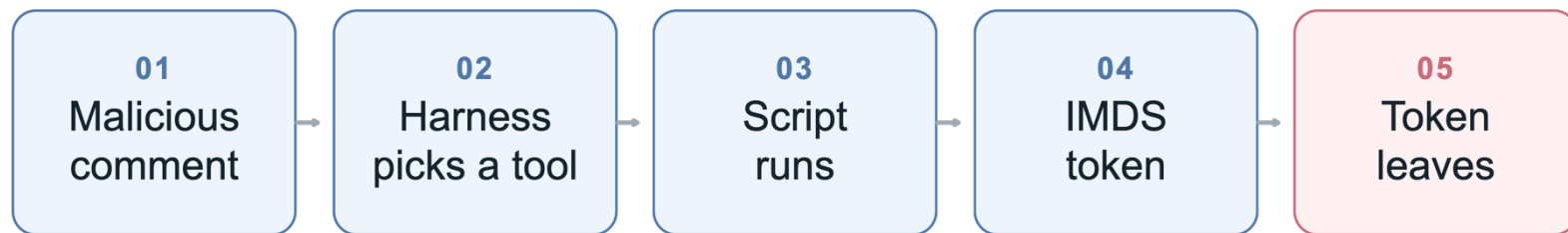


02

Why outside the harness

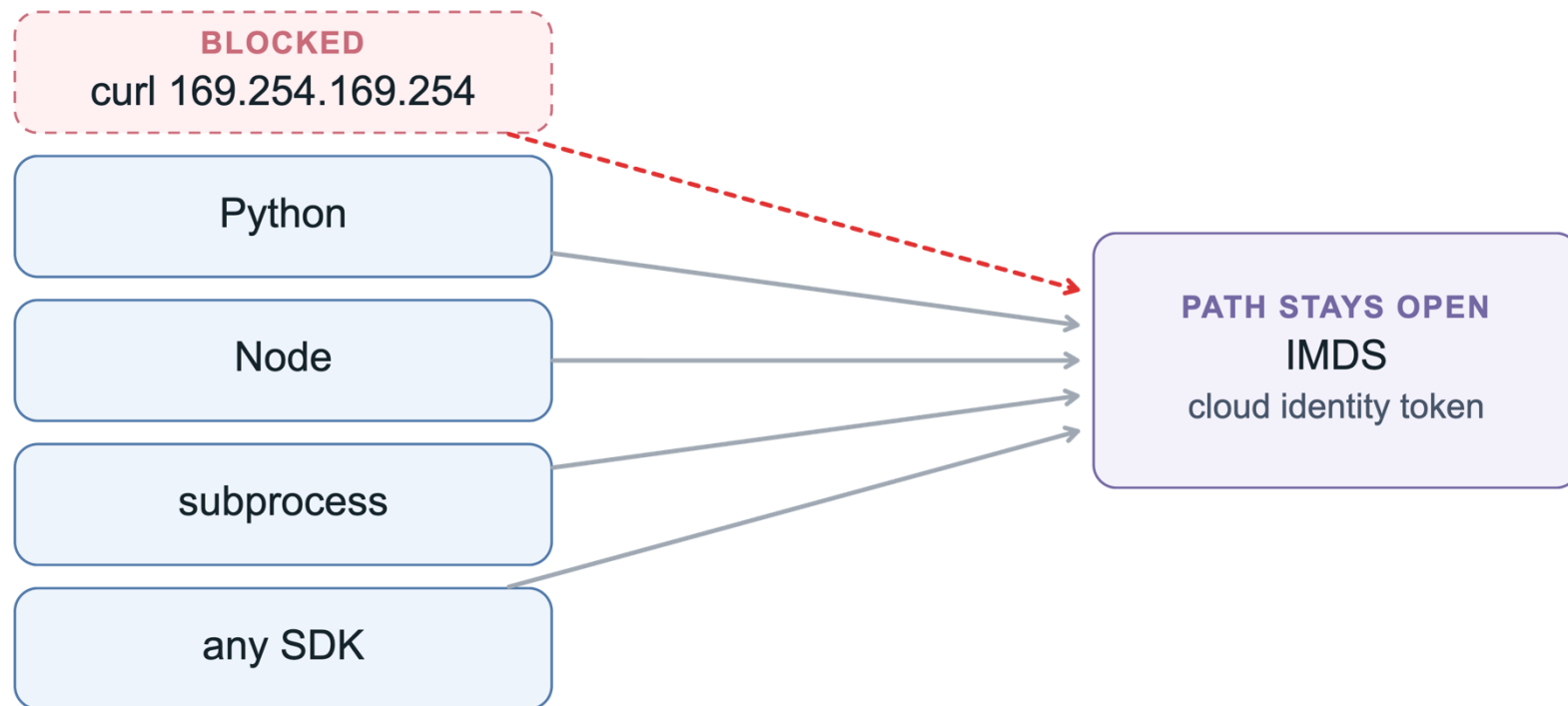


Attack chain



No escape · no alert · Pod stays healthy

Failed fix



Runtime boundary

Kubernetes compute · storage · orchestration

Agent Runtime identity · network · budget · audit

Agent Instance Kata VM sandbox · runs code

Agent Harness
plans · picks tools · MCP · Skills



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03

Lessons



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Boundary

INSTANCE

Kata VM sandbox

untrusted code
processes · files · resources

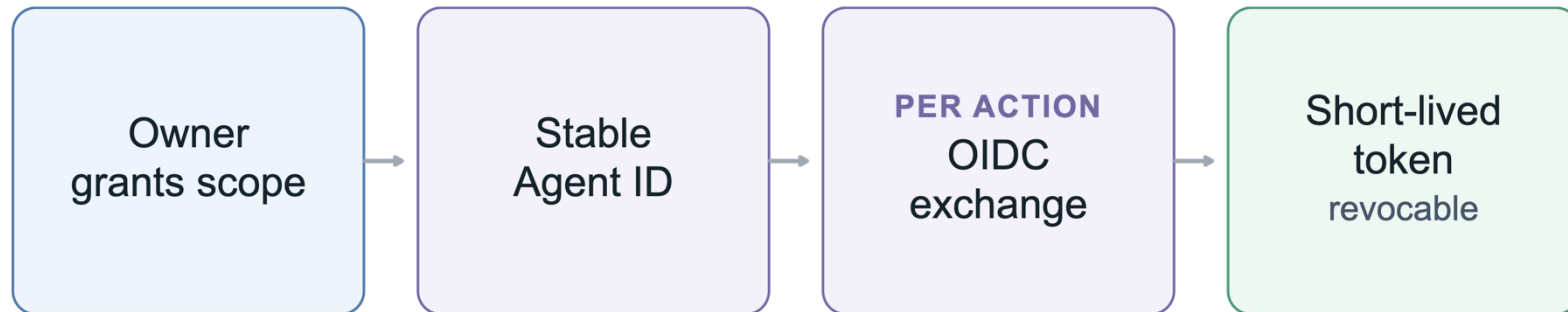
OUTSIDE THE SANDBOX

Runtime keeps
identity · network · budget · audit

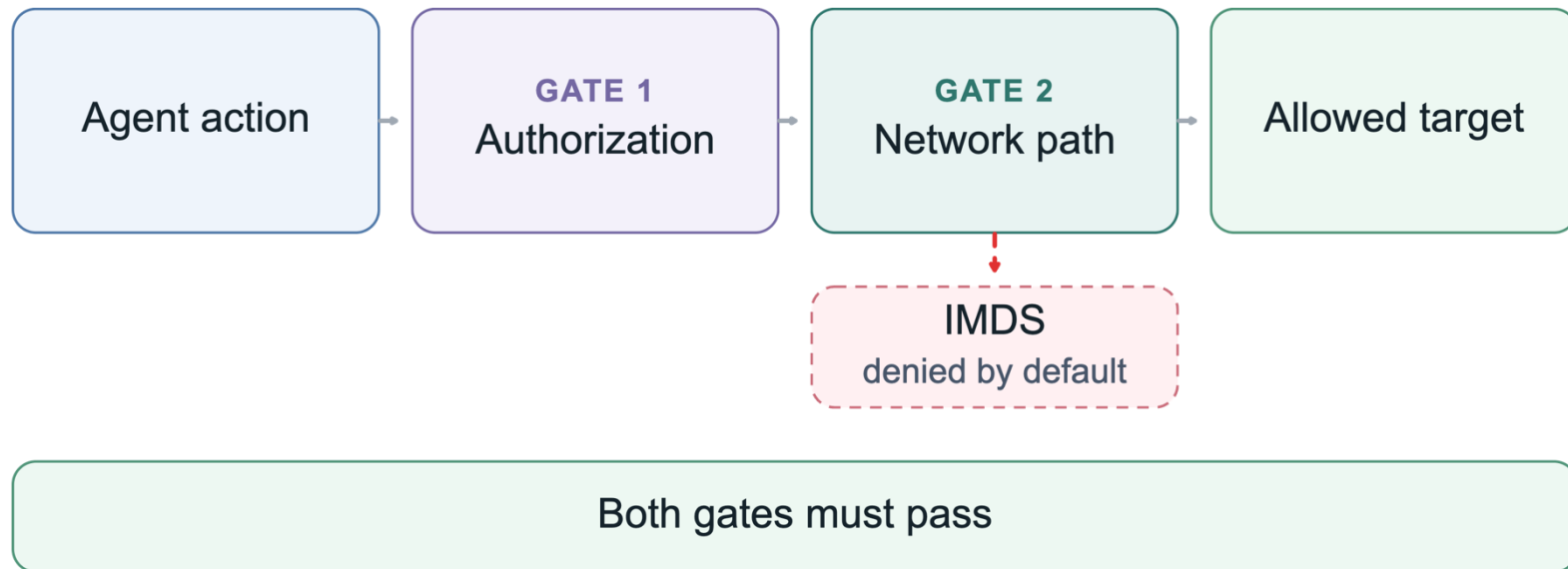
Cost

slower boot · lower density · compatibility checks

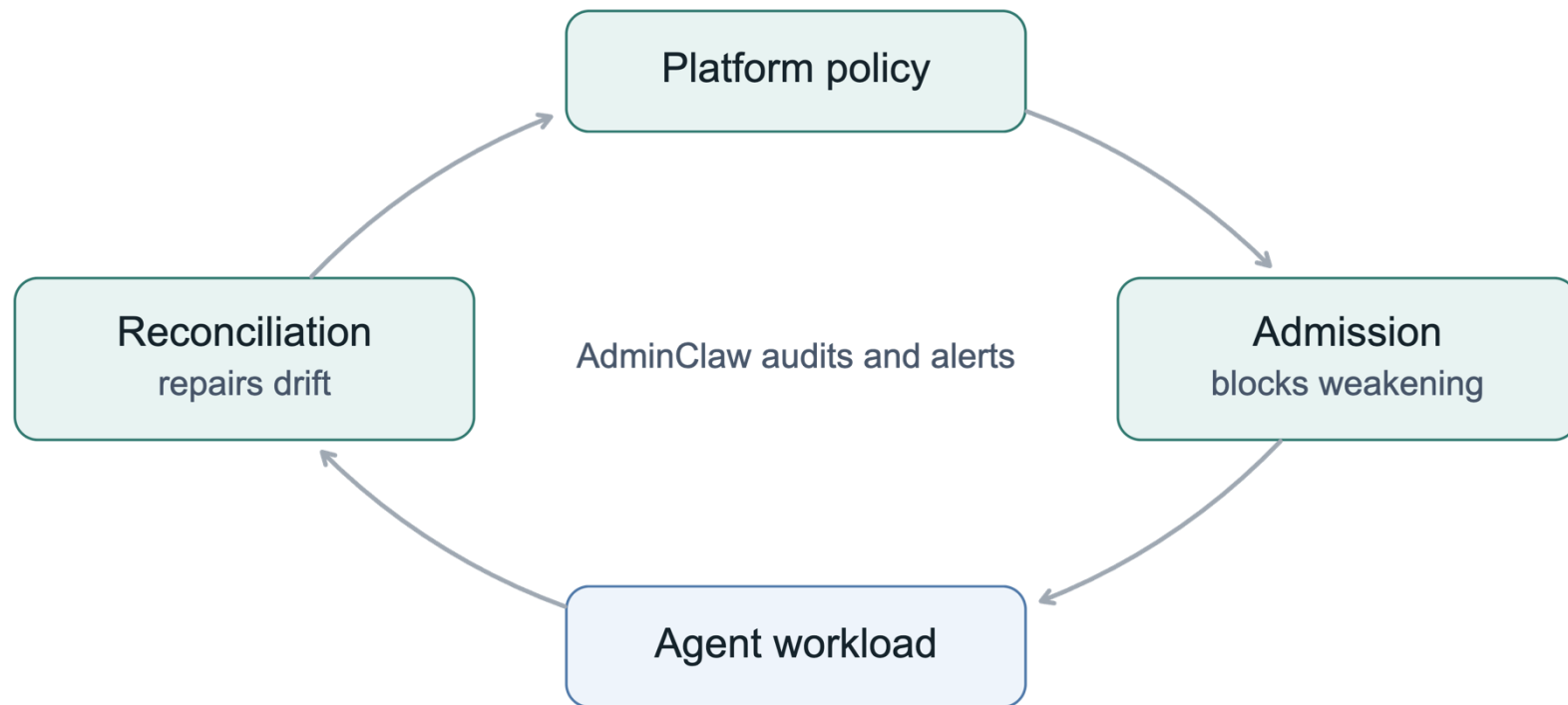
Identity



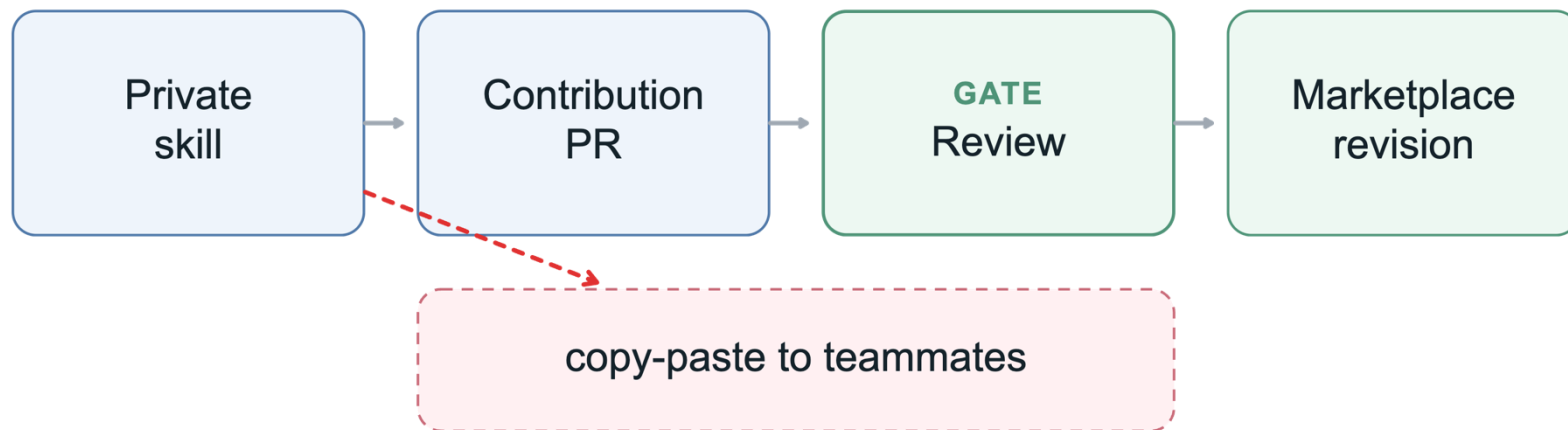
Reach



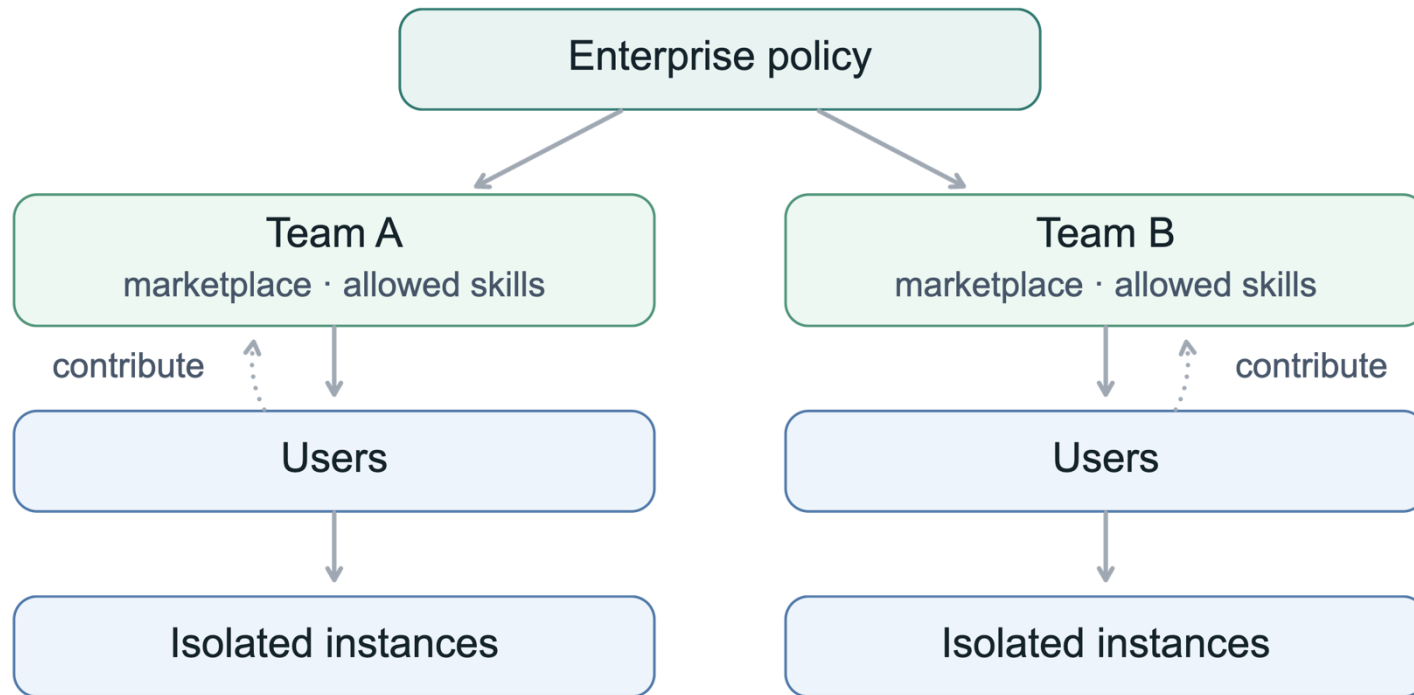
Reach



Capability



Capability



Scale

FLEET WIDTH

more users
more teams

TIME

long runs
replaced compute

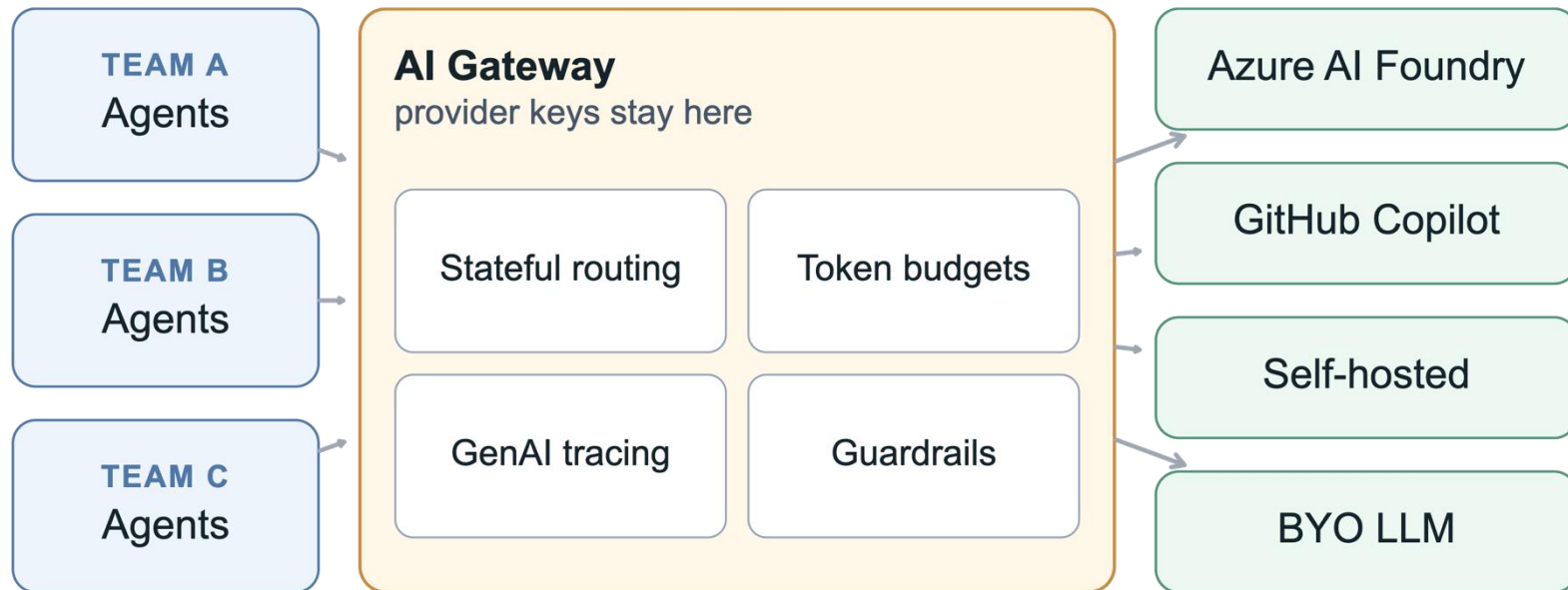
DELEGATION

parent run
child runs

Same invariants

identity · revision · budget · audit

Scale



Roadmap

DELIVERED

Agent Runtime foundation

sandbox · identity · controlled access · AI gateway · skills · governance

PLANNED

Pluggable harness

bring your own agent, configured per user

Pluggable sandbox

MicroVM runtimes beyond Kata

Finer identity and permissions

different scopes for different tasks

Local model access

inference models running in the same cluster

04

Takeaways



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Takeaways

1 The harness is an input, not a boundary

A control the agent can change is not a boundary, only a layer of defence

2 A boundary holds regardless of how the agent acts

Sandbox, per-action identity, network path: enforced by the platform, invisible to the harness

3 Design for the agent that outlives the engineer

Sessions survive pods, skills carry revisions, budgets stop hard at the gateway

Q&A

